

SUBMISSION 1 — SYNOPSIS / PROJECT PROPOSAL

Collaborative Study Material & Quiz Platform

1. Project Identification

Project Title	Collaborative Study Material & Quiz Platform
Group Number	1
Team Number	Team 1
Course	UE24CS341A – Software Engineering
Semester	V Semester
Development Methodology	Agile (iterative, sprint-based)
Complexity	Medium
Technology	MERN Stack (MongoDB, Express, React, Node.js) — primary stack
Customer / Stakeholder	Students and educators; course faculty as evaluating stakeholder
Version	1.0

2. Team Profile & Product Ownership

Member	SRN	PRN	Owned Functionality	Secondary Responsibilities
Anisha Nukul Niranjan	PES1UG24CS064	PES1202402163	User accounts & profiles — registration, login, role-based profiles (student / educator / admin)	API review, authentication support
Aryaroop Bose	PES1UG24CS087	PES1202400286	Study material management & moderation — upload, tag, categorize, search, share, collaborative curation & reporting/moderation support	UI / UX
Ayush Ravindra Paithankar	PES1UG24CS104	PES1202400033	Quiz creation & management — build quizzes, question types, answer keys, edit / publish; plus performance leaderboard	Database design
Bharat V Bhat (Team Lead)	PES1UG24CS113	PES1202403301	Quiz attempts, scoring & progress analytics — attempt flow, auto-scoring, results history & progress dashboard	Testing / CI, coordination

3. Problem Statement

Students preparing for courses rely on study material and self-assessment, but today these are fragmented. Notes and resources are scattered across chats, drives and personal folders, there is no shared, organized repository, and quizzing for self-assessment usually happens through ad-hoc or paid tools that are disconnected from the study material itself. As a result, students struggle to find quality material, cannot easily test their understanding of a specific topic, and have no visibility into their own progress over time.

4. Proposed Project Description

The proposed project, the **Collaborative Study Material & Quiz Platform**, is a web application that brings study material and self-assessment together in one place. Registered users (students and educators) can access study material, while students with contributor privileges and educators can upload study material with a title, category and tags, then search, browse and share it, and collaboratively curate it through ratings, comments and a simple revision history. Students with contributor privileges and educators can create topic-linked quizzes with multiple question types and defined answer keys; students can attempt published quizzes and receive automatic scoring with immediate per-question feedback. Each student can track performance over time through a progress dashboard and a performance leaderboard whose rankings are derived from stored quiz-attempt scores. Access is role-based and secured, and the product is built incrementally on the MERN stack using Agile practices — keeping learning and assessment in one connected place.

5. Objectives

- Provide a single, organized, searchable repository for shared study material.
- Enable collaborative curation of material through tagging, sharing, ratings and comments.
- Allow users to create and manage quizzes with multiple question types linked to topics.
- Let students attempt quizzes with reliable automatic scoring and immediate feedback.
- Give students visibility into their performance through a progress dashboard and performance leaderboard.
- Enforce secure, role-based access for students, educators and administrators.
- Develop the product incrementally using Agile practices with evidence of each stage.

6. Target Users / Stakeholders

User / Stakeholder	Goal / Need	Characteristics
Student (primary)	Find material, self-assess with quizzes, track progress	Frequent user, non-technical, peak use before exams
Contributor (a capability of a student, not a separate account role)	Uploads, organizes and curates shared material and creates quizzes	Any student who actively contributes; helps maintain content quality
Educator	Share authoritative material and create assessments	Higher-privilege, trusted content author
Administrator	Manage users and role/privilege assignment; moderate reported content	Maintains platform health, security and content moderation
Course faculty (evaluator)	Assess SDLC process and deliverables	Reviews documentation, code and demo

7. Scope

In Scope	Out of Scope
User registration, login & role-based profiles	Real-money payments / subscriptions
Study material upload, tagging, categorization & search	Automated AI content generation of study material
Collaborative curation (sharing, ratings, comments)	Live video classes / real-time streaming
Quiz creation & management (multiple question types)	Institution-wide grade / ERP integration
Quiz attempt with automatic scoring & feedback	AI-based automatic essay/subjective grading
Progress analytics dashboard & performance leaderboard	Native mobile apps (web-responsive only)
Role-based access control & input validation	Multi-institution enterprise deployment

8. Functional Features

1. User registration, login and role-based profiles (student / educator / administrator).
2. Study material upload with tagging and categorization.
3. Search, filter and browse study material by topic / tag.
4. Collaborative curation — sharing, ratings, comments and simple revision history (tracking edits made to material).
5. Quiz creation and management with multiple question types and answer keys.
6. Quiz attempt with automatic scoring and immediate feedback.
7. Results history and progress analytics dashboard.
8. Performance leaderboard (rankings derived from quiz-attempt scores) to encourage participation.

9. Short-Term Plan (Agile Sprints)

Iteration	Planned Output	Owner	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, authentication & profiles	All / Anisha	User stories + Git repo + working login
Sprint 2	Study material upload, tagging, categorization & search	Aryaroop	Demo + unit tests
Sprint 3	Collaborative curation — sharing, ratings, comments	Aryaroop / All	Integration tests + PRs
Sprint 4	Quiz creation & management	Ayush	System demo + test evidence
Sprint 5	Quiz attempt, scoring & progress analytics; leaderboard & security hardening	Bharat / Ayush	Validation + coverage report
Sprint 6	System Integration, Final Validation & Release	All	Final report + deployment / demo

10. Proposed Technology / Tool Stack

Activity	Tool / Technology	Evidence to Produce
Frontend	React (MERN)	Responsive web UI
Backend	Node.js + Express	REST API
Database	MongoDB	Schema / collections
Version control	Git + GitHub	Repo URL, commit history
Agile planning	Jira / GitHub Projects	Backlog, user stories, sprint board
CI/CD	Jenkins	Pipeline logs / screenshots
Containerization	Docker	Dockerfile + build evidence
Testing	Jest (JavaScript unit / integration tests)	Test + coverage reports
Static analysis	SonarQube	Analysis report
API testing	Postman	Collection / results
Design / modeling	Draw.io / StarUML / PlantUML	UML diagrams

11. Agile Justification

Agile suits this project because the platform is naturally divisible into independent, incrementally deliverable features (accounts, material management, quizzes, analytics), each owned by a team member. Sprint-based development lets us deliver a working slice every iteration, gather feedback, and continuously integrate and test rather than integrating everything at the end. User stories, a maintained backlog, sprint boards and per-member task ownership will provide continuous evidence of individual contribution, in line with the course requirements for GitHub/Jira activity, code reviews, CI/CD and security validation.